

Synchronised Multimedia Integration Language

The need for Web-based applications is growing and Synchronised Multimedia Integration Language (SMIL) is a great tool that allows Web developers to streamline huge audio-video applications in a lighter format on the Web. SMIL, pronounced as 'smile', sends the images, files, text and sound in a structured format. This means that though the files may be at different locations, their timing would be perfect when embedded in an application and would not hamper the speed of display on the Net, enabling the image to download faster. SMIL is the current standard incorporated in most browsers these days to enhance Web applications.

Learning SMIL is easy, as it is based on XML. It is in fact a mark up language like HTML and the coding procedure is simple and easy to understand and can be written using a simple text editor such as Notepad.

Its best utility can be seen in real audio and real video that work with Real player. For instance, you can pick a Flash file, convert it to SMIL and play it in a media player. It is a versatile tool for Web-based applications, especially when bandwidth is the major concern.



Hands-on

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as compared to Director or Authorware. It uses Action Scripting for defining scene sequences, form inputs and other behaviour that are required to jump start other programme interfaces. The newer version, Flash 5, incorporates a fair bit of built-in scripting elements.

Where do you start?

Learning languages such as C, C++, Visual C++, Java and VB script should give them a good jump start in multimedia programming, as they form the programming core. These languages, coupled with Director, Authorware and Flash, are widely used for developing interactive CD-ROMs and Web pages. Knowledge of a visual programming tool such as Visual Basic or Power Builder (PB) would be an added advantage as it smoothen the transition to learning Director.

The same is true for packages such as Flash and Dreamweaver, which allow you to view the code and make changes accordingly. Besides, knowledge of 3D modelling tools such as 3D Studio Max and Elastic Reality are a must in today's multimedia applications.

Where can you learn?

Owing to the presence of a large market and a huge potential waiting to be tapped, there are several institutes that offer courses in multimedia programming and design. Major players such as Pentamedia and Arena Multimedia not only develop multimedia applications using state-of-the-art infrastructure and resources, but also offer training courses in this field.

Depending on which institute you choose, you could learn modules ranging from the very basic to high-end development applications; from designing Web pages to Web server application development such as ASP, JSP and PHP. Arena Multimedia (www.arena-multimedia.com) offers good infrastructure and caters to almost all aspects of working with multimedia, which covers basic programming concepts of multimedia for graphic visualisers/specialists, Web page developers, and also covers 3D animation and high-end multimedia programming. Others like Edit Institute (www.editinstitute.com), Microcell India (www.microcellindia.com) and the Workstation (www.tasaa.com) also cater to these segments.

You can also learn on your own as there are many CBT titles now available in the market. New methods and models are being deployed to make the learning curve easy and interactive, by making self-explanatory and user-friendly tutorials. Pentamedia claims to have about 170 titles

ranging from subjects related to IT, medicine and other informative topics.

Keep in mind, however, that though there are several institutes that would teach multimedia programming, it is up to you to show your versatility. It is advisable to keep your portfolio ready as that is your passport to success. The projects must reflect the kind of work you've been involved with and the diversified range of programming applications you are familiar with.

However, when it comes to judging an individual's potential, your portfolio will no doubt come in handy, but the usual practice is that companies will give you a small multimedia assignment (called real time testing) that you need to finish in a certain time. Companies do this because it is easy to generate code, modify it and pass it off as your own work. "This concept of real time testing is very handy as it helps the employer in judging the calibre of the candidate.



The European markets are opening up and there is a good enough demand for multimedia professionals

Rajiv Sangari

Deputy General Manager, Pentamedia Graphics

With this, one can make out whether the candidate is strong in his fundamentals and can think clearly on the project given," adds Kiran Nayak. Moreover, while on the job, the programmer must be able to grasp the fundamentals easily and develop new techniques to improve work efficiency. Efficient use of libraries and templates go a long way in building effective source code. The programmer must also be able to judge on his own as to which source code would be the best for writing a program. Termed as 'smart programming' in a programmers dictionary, this is what makes a programmer stand out from the rest of the pack.

The opportunities

Now that you are armed with the requisite skills, where do you go? Well, opportunities abound for multimedia programmers in areas such as television, advertising, filmmaking, content development, and of course the Internet.

The advertising field has room for everyone, from graphic designers to layout artists to creative directors. You could start off as a designer or visualiser and grow to become an art and media consultant, studio manager or even a creative director.

The film and TV industry are the hotspots